



## Montse Fuentes

St. Edward's University

[→About Montserrat Fuentes](#)

Tuesday, October 6, 2026

2:00 pm – 3:00 pm

Location: Long Hall 216

## Spatial Statistical Modeling of Neuroimaging Data and Drug Addiction

---

### Abstract

---

High-resolution neuroimaging data provide unprecedented opportunities to study brain structure and function, but their massive dimensionality and complex spatial dependence create fundamental statistical challenges. Magnetic resonance imaging (MRI), for example, may contain millions of spatially correlated voxels. Fully modeling this dependence is computationally prohibitive, leading many approaches to aggregate information into predefined regions of interest (ROIs). Although convenient, such aggregation can obscure fine-scale spatial structure, diminish power to detect localized effects, and potentially alter scientific conclusions.

We introduce a novel Bayesian tensor regression framework for modeling neuroimaging data directly at the voxel level, avoiding the need to collapse the brain into predefined regions. The brain image is represented as a tensor-valued response, allowing covariate effects and spatial dependence to be modeled across its natural multidimensional structure. A generalized sparsity formulation makes inference feasible at very high dimensions while preserving scientifically meaningful spatial information. The fully Bayesian framework also provides coherent quantification of uncertainty through computational algorithms designed for large-scale tensor data.

We apply the methodology to neuroimaging studies of drug addiction, with particular emphasis on cocaine dependence and its effects on brain structure and function. Simulation studies and empirical analyses demonstrate improved identification of spatially localized patterns and a richer characterization of brain alterations associated with addiction. More broadly, this work shows how spatial statistics can move neuroimaging analysis beyond coarse regional summaries toward scalable, uncertainty-aware inference at the resolution at which brain changes are actually observed.

### About the Speaker

---

Montserrat Montse Fuentes is a distinguished statistician and academic leader whose research focuses on spatial and spatio-temporal statistics, with applications in atmospheric science, climate, and environmental health. She served as President of St. Edwards University from 2021 to 2026 and previously as Executive Vice President and Provost at the University of Iowa. Earlier in her career, she held faculty and leadership positions at North Carolina State University, including Head of the Department of Statistics. Fuentes is a Fellow of the American Statistical Association, the Institute of Mathematical Statistics, and the American Association for the Advancement of Science.